Velocity and Acceleration of VVF

section 12.3

9-21-26

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11) a) $v(t) = \langle 1, 2t, t \rangle$
$a(t) = \langle 0, 2, 1 \rangle$
$\|v(t)\| = \sqrt{1+5t^2}$

b) $v(4) = \langle 1, 8, 4 \rangle$
$a(4) = \langle 0, 2, 1 \rangle$

13) a) $v(t) = \langle 1, 1, \frac{-\cancel{2}t}{\cancel{2}\sqrt{9-t^2}} \rangle$
$a(t) = \langle 0, 0, \frac{-9}{(9-t^2)^{3/2}} \rangle$

$\|v(t)\| = \frac{\sqrt{18-t^2}}{\sqrt{9-t^2}}$

b) $v(0) = \langle 1, 1, 0 \rangle$
$a(0) = \langle 0, 0, -\frac{1}{3} \rangle$

17) a)

$v(t) = \langle e^t\cos t - \sin t\, e^t, e^t\sin t + e^t\cos t, e^t \rangle$
$\|v(t)\| = e^t\sqrt{(\cos t-\sin t)^2+(\sin t+\cos t)^2+1}$
$= e^t\sqrt{2\cos^2 t + 2\sin^2 t + 1}$

$a(t) = \langle -2e^t\sin t, 2e^t\cos t, e^t \rangle$

b)

$v(0) = \langle 1, 1, 1 \rangle$
$a(0) = \langle 0, 2, 1 \rangle$

21) $a(t) = \langle 0, t, t \rangle$
$v(t) = \langle c_1, \frac{t^2}{2}+c_2, \frac{t^2}{2}+c_3 \rangle$
$v(1) = \langle 0, 5, 0 \rangle$
$v(t) = \langle 0, \frac{t^2}{2}+\frac{9}{2}, \frac{t^2}{2}-\frac{1}{2} \rangle$
$r(t) = \langle c_1, \frac{t^3}{6}+\frac{9}{2}t+c_2, \frac{t^3}{6}-\frac{t}{2}+c_3 \rangle$
$r(1) = \langle 0, 0, 0 \rangle$
$r(t) = \langle 0, \frac{t^3}{6}+\frac{9}{2}t-\frac{28}{6}, \frac{t^3}{6}-\frac{t}{2}+\frac{2}{6} \rangle$
$r(2) = \langle 0, \frac{17}{3}, \frac{2}{3} \rangle$

23) $a(t) = \langle -\cos t, -\sin t, 0 \rangle$
$v(t) = \langle -\sin t + c_1, \cos t + c_2, c_3 \rangle$
$v(0) = \langle 0, 1, 1 \rangle$

$v(t) = \langle -\sin t, \cos t, 1 \rangle$

$r(t) = \langle \cos t + c_1, \sin t + c_2, t + c \rangle$

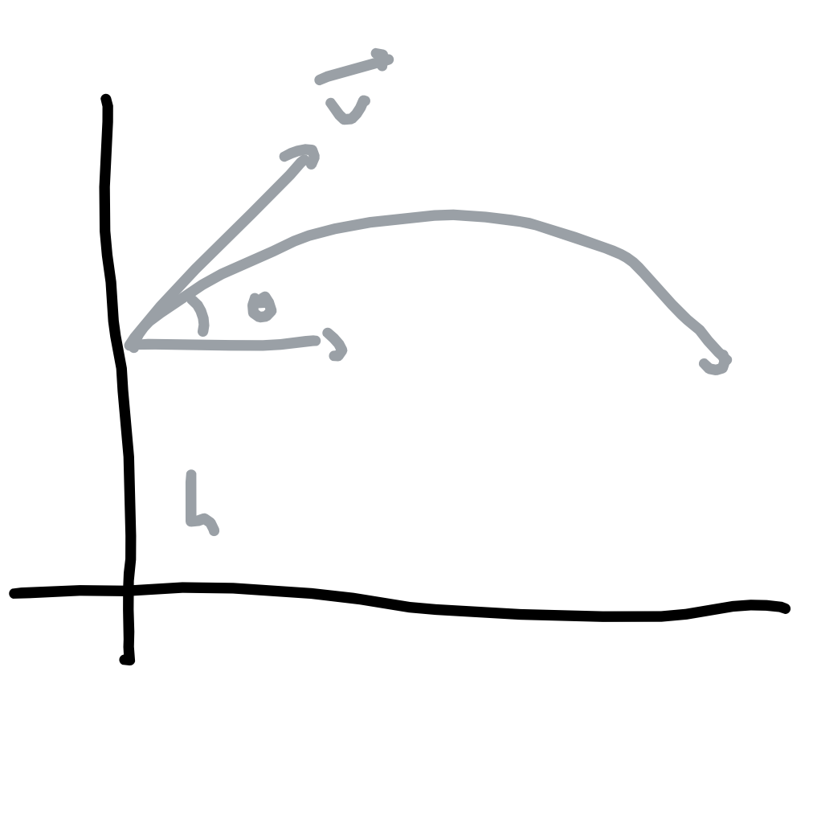
$r(0) = \langle 1, 0, 0 \rangle$

$r(t) = \langle \cos t, \sin t, t \rangle$

$r(2) = \langle \cos 2, \sin 2, 2 \rangle$

B) Projectile Motion

$x(t) = V_0 t\cos\theta$
$y(t) = -\frac{g}{2}t^2 + V_0 t\sin\theta + h$



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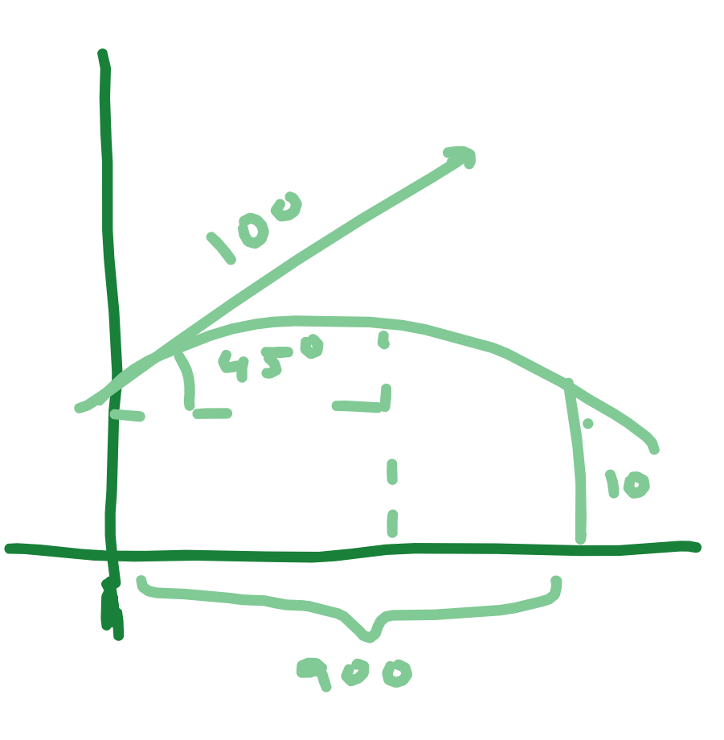
ex) A baseball is hit 3ft above the ground at 100ft/sec with an ∡ of 45°

$x(t) = 50\sqrt{2}(t)$
$y(t) = -16t^2 + 50\sqrt{2}t + 3$



a) find the max height

$t = \frac{50\sqrt{2}}{32}$

$-16\left(\frac{50\sqrt{2}}{32}\right)^2 + \frac{50\sqrt{2}(50\sqrt{2})}{32} + 3$

$\frac{(50\sqrt{2})^2}{32}\left(-\frac{1}{2}+1\right) + 3$

b) will it clear a fence 10ft away?

$\frac{300}{50\sqrt{2}} = 50\sqrt{2}t$

$\frac{6}{\sqrt{2}} = t =$

$-16(3\sqrt{2})^2 + 50$

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27) $\langle V_0\frac{\sqrt{2}}{2}t, -16t^2 + V_0\frac{\sqrt{2}}{3} + 3 \rangle$

$300 = \frac{V_0}{\sqrt{2}}t$ $\quad -16t^2 + \frac{V_0}{\sqrt{2}}t + \cancel{3} = \cancel{3}$

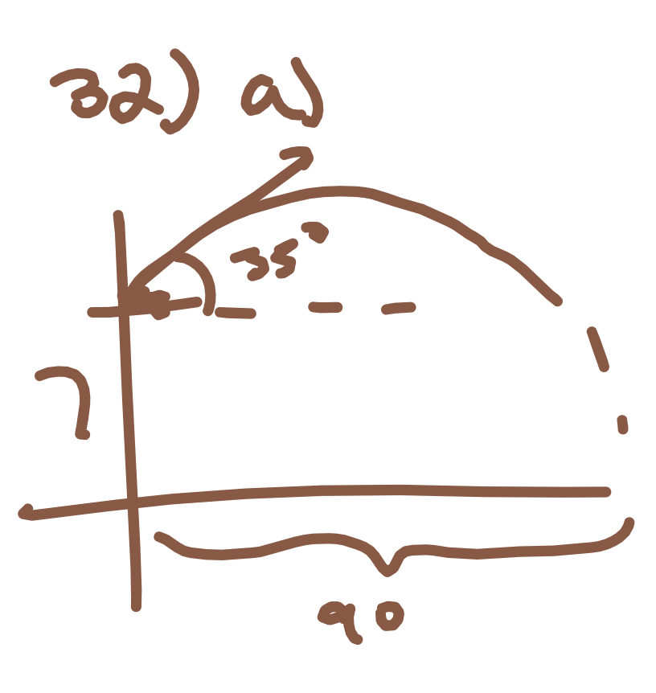
$300\sqrt{2} = V_0 t$ $\quad 16t^2 = 300$

$\frac{300\sqrt{2}}{t}$ $\quad t = \sqrt{\frac{300}{16}}$

$\frac{300\sqrt{2}}{\sqrt{300/16}} = \frac{300\sqrt{2}}{\frac{10\sqrt{3}}{4}}$

$\frac{1200\sqrt{2}}{10\sqrt{3}} = \boxed{40\sqrt{6}}$

32) a)

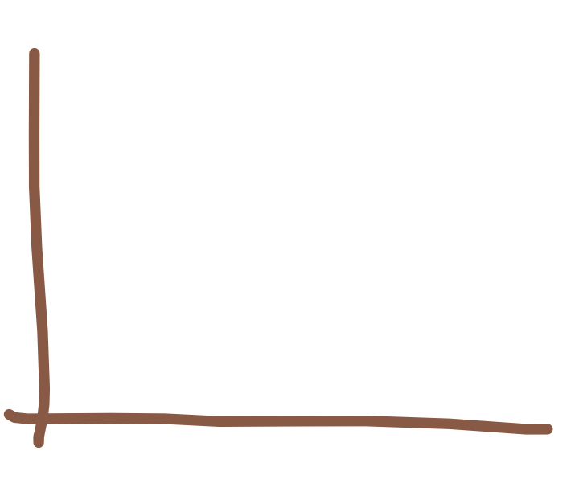


$\overset{90}{\cancel{x(t)}} = V_0\cos 35\cdot t$
$\frac{90}{\cos 35} = V_0\cdot t$

$x(t) = -16t^2 + V_0\sin 35t + 7$
$4 = -16t^2 + V_0\sin 35t + 7$
$0 = -16t^2 + \left(\frac{90}{\cos 35\cancel{t}}\right)\sin 35\cancel{t} + 3$
$t = \sqrt{\frac{90\tan 35}{16}}$

$V_0 = \frac{90}{t\cos 35}$

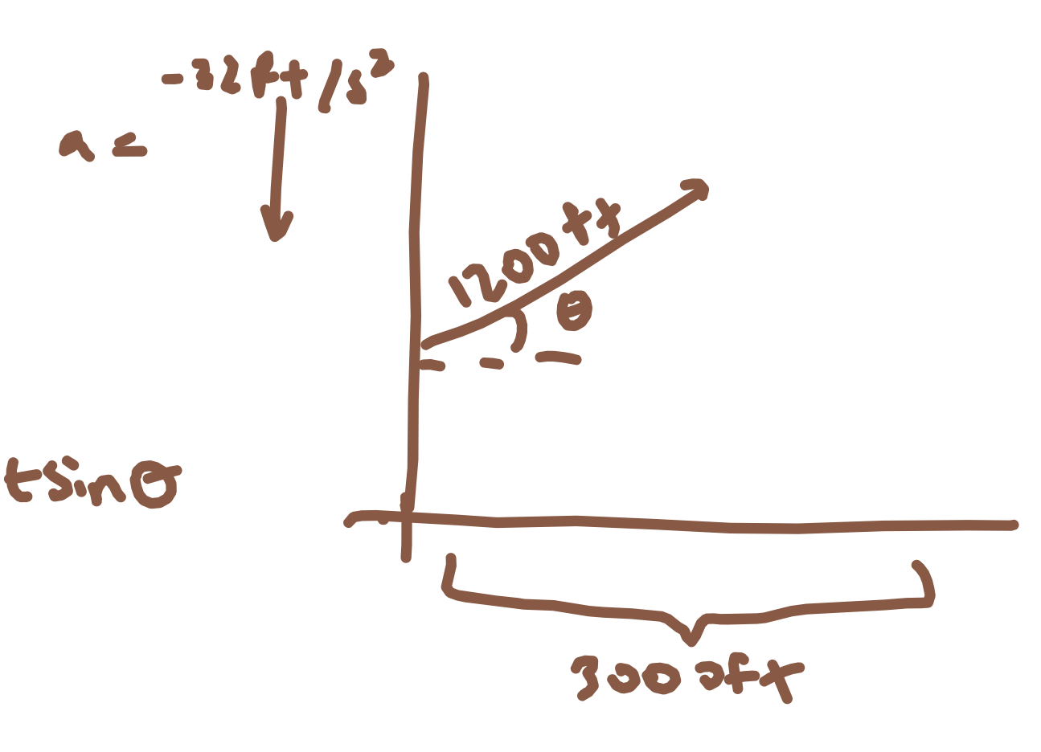
34) $x(t) = 792t$
$y(t) = -16t^2 + 30000$



$r(t) = \langle 792t, -16t^2 + 30000 \rangle$
$r'(t)$: $v(t) = \langle 792, -32t \rangle$
$v'(t) = 0$
$t = 25\sqrt{3}$
$x(25\sqrt{3}) = 19800\sqrt{3}$
$\|v(25\sqrt{3})\| = 8\sqrt{39801}$
$\tan\theta = \frac{30000}{19800\sqrt{3}} = \frac{50\sqrt{3}}{99}$

$\theta = 41.1786$ below the horizontal

35)

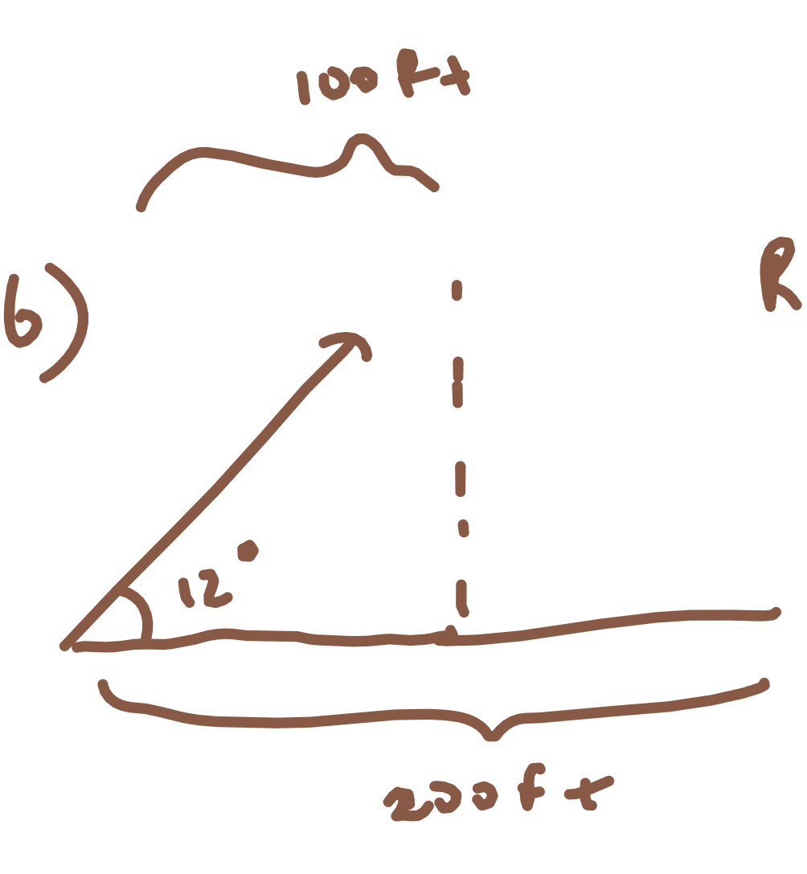


$x(t) = 1200t\cos\theta$
$y(t) = -16t^2 + 1200t\sin\theta$

$3000 = 1200t\cos\theta$
$\frac{3000}{1200\cos\theta} = t$

$x = \frac{1200^2}{32}\sin(2\theta) = 3000 \Rightarrow \sin(2\theta) = \frac{1}{15}$
$\theta = 1.91$

36)



$R = \frac{V_0^2\sin(2\theta)}{g}$
$V_0 = \sqrt{\frac{R\cdot g}{\sin(2\theta)}}$
$\sin(24°) = 0.406737$
$V_0 = 125.439ft/s$